Hybrid Gated Fusion for SaaS Sales Outcome Prediction: Combining Tabular Features and Deep Text Representations

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***Abstract*—This paper presents Phase 3 of the Predictive Sales Analytics Engine, which unifies structured tabular features with learned deep text representations via a learnable scalar fusion gate. Operating on 100,000 SaaS sales conversations, the full hybrid model achieves 97.61% F_1 and 99.81% ROC-AUC—surpassing Phase 1 Logistic Regression (96.39% F_1, 99.57% AUC) while attaining 86.27% edge-case F_1 on hard-to-classify borderline samples. Six ablation variants validate every architectural component: gated fusion outperforms simple concatenation ($\Delta F_1 = +0.73\%$), XGBoost leaf encoding raises edge-case robustness by $+11.48\%$, and freezing the DistilBERT body yields equivalent performance at lower training cost. A cross-phase comparison against Phase 1 (Logistic Regression, Random Forest) and Phase 2 (BiLSTM, GloVe BiLSTM) demonstrates that complementary modalities fused through a learned gate consistently outperform any single-modality pipeline.**

***Index Terms*—hybrid fusion, gating mechanism, DistilBERT, XGBoost, transfer learning, sales outcome prediction, ablation studies, edge-case robustness, NLP.**

I. Introduction

Phase 1 of the Predictive Sales Analytics Engine established strong classical ML baselines—96.34% accuracy and 99.57% ROC-AUC—using ℓ_2-regularised Logistic Regression on hand-engineered engagement features and pre-computed embeddings [1]. Phase 2 replaced that pipeline with end-to-end deep learning (BiLSTM, GloVe-initialised BiLSTM, DistilBERT), achieving 83.89% accuracy on raw text alone but consistently underperforming Phase 1 due to the absence of structured domain signals.

Phase 3 resolves this gap with a central insight: tabular features and text representations are *complementary*, not competing. A learnable scalar gate dynamically interpolates between both modalities per sample, allocating higher weight to structured signals when conversational text is short or noisy, and to text representations when linguistic cues are predictive.

A. Objectives

Phase 3 pursues four goals: (1) design a fusion architecture connecting XGBoost leaf encodings to a DistilBERT text branch via a learned gate; (2) validate each component through six controlled ablation variants; (3) quantify edge-case robustness as a distinct evaluation axis beyond aggregate accuracy; and (4) produce a unified cross-phase comparison on the same 20,000-sample held-out test set used across all prior phases.

B. Key Contribution: Learnable Gated Fusion

Rather than naively concatenating two representation vectors, Phase 3 learns a scalar gate $g \in [0, 1]$ combining the 256-d tabular projection $\mathbf{z}_{\text{tab}}$ and text projection $\mathbf{z}_{\text{text}}$ as

$$\mathbf{z}_{\text{fused}} = g \cdot \mathbf{z}_{\text{text}} + (1 - g) \cdot \mathbf{z}_{\text{tab}} \tag{1}$$

where $g = \sigma\big(\mathbf{w}_g^\top [\mathbf{z}_{\text{text}};\, \mathbf{z}_{\text{tab}}] + b_g\big)$ is conditioned on both representations jointly, allowing the model to calibrate modality trust dynamically during training.

II. Architecture

A. Tabular Branch: XGBoost Leaf Encoding

The trained XGBoost forest from Phase 1 serves as a deterministic feature extractor. Each input sample traverses every decision tree, producing a leaf-node index per tree. These indices are one-hot encoded and concatenated into a sparse binary vector (≈100–200 dimensions), then projected to 256-d via a fully connected layer with ReLU activation and 50% dropout. This encoding captures non-linear feature interactions and thresholds learned in Phase 1 without retraining the forest.

B. Text Branch: DistilBERT Encoder

Conversation text is tokenised with the HuggingFace `distilbert-base-uncased` WordPiece tokeniser (max 512 tokens). The DistilBERT body (6 Transformer layers, 768-d hidden, ≈66M parameters) produces contextualised token embeddings. The `[CLS]` token representation—a global summary of the conversation after six layers of multi-head self-attention—is extracted and passed through LayerNorm (768-d), 30% dropout, and a linear projection ($768 \rightarrow 256$),

TABLE I: Training hyperparameters for the full hybrid model.

Hyperparameter	Value
Optimiser	AdamW
LR — classification head	2×10^{-4}
LR — DistilBERT body	2×10^{-5}
Weight decay	1×10^{-2}
Batch size	32
Max epochs	8
Loss function	CrossEntropyLoss
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 1)
Early stop patience	2 epochs, $\delta = 10^{-4}$
Gradient clip norm	1.0
Dropout (text & fusion)	30%
Dropout (tabular projection)	50%

yielding $\mathbf{z}_{\text{text}}$. Differential learning rates are applied: the body at 2×10^{-5} and the projection head at 2×10^{-4}, preventing catastrophic forgetting while adapting higher-level layers to SaaS conversation patterns [2], [3].

C. Fusion and Classification Head

Both branches produce 256-d representations, combined via Eq. (1). The resulting fused vector is normalised with LayerNorm, regularised with 30% dropout, and fed to a linear classification head ($256 \rightarrow 2$, Kaiming initialisation) trained with CrossEntropyLoss.

D. Training Configuration

Table I summarises hyperparameters. AdamW with decoupled weight decay [**?**] is applied throughout. Early stopping monitors validation F_1 with patience of two epochs, and `ReduceLROnPlateau` halves the learning rate when validation loss stagnates for one consecutive epoch.

III. Experimental Results

A. Full Hybrid Performance

The full hybrid (Variant A) achieves 97.67% accuracy, precision 0.9793, recall 0.9735, F_1 0.9761, ROC-AUC 0.9981, average precision 0.9982, and edge-case F_1 0.8627 on the 20,000-sample test set. This surpasses Phase 1's strongest baseline—Logistic Regression at 96.34% accuracy, F_1 0.9639, AUC 0.9957—while delivering meaningful robustness improvements on borderline conversations.

B. Ablation Study

Six controlled variants systematically isolate the contribution of each architectural component. Table II reports all metrics alongside a precise description of what each variant modifies relative to the full hybrid.

Fig. 1 visualises all six variants across four metrics simultaneously. The full hybrid (A) leads on overall F_1 (0.9761) and edge-case F_1 (0.8627). Replacing the gate with simple concatenation (D) drops F_1 by -0.73% and edge-case F_1 by -1.47%, confirming that learned modality weighting matters beyond static merging. Removing XGBoost leaf encodings (F) collapses edge-case F_1 by -11.48% while only reducing

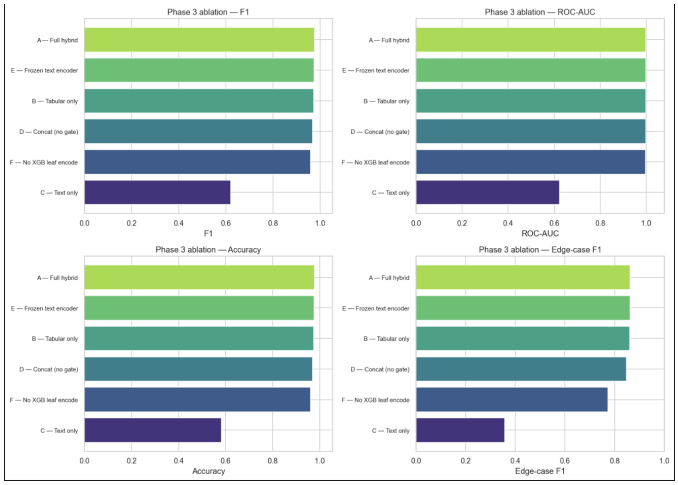

Fig. 1: Phase 3 ablation across six variants on four metrics: F_1 (top-left), ROC-AUC (top-right), Accuracy (bottom-left), and edge-case F_1 (bottom-right). The full hybrid (A) leads across all four. Variant C (text only) fails catastrophically, confirming the necessity of tabular signals.

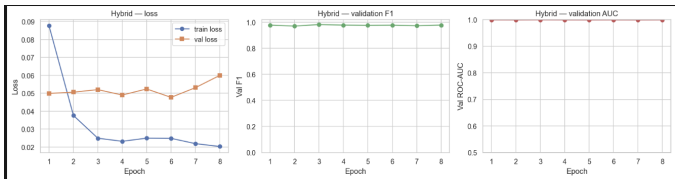

Fig. 2: Full hybrid training dynamics over 8 epochs. Cross-entropy loss decreases monotonically (left); validation F_1 (centre) and ROC-AUC (right) plateau near their ceiling within two epochs, indicating rapid and stable convergence.

overall F_1 by -1.68%—revealing that tree-derived features are disproportionately important for hard samples. Freezing the DistilBERT body (E) costs only -0.14% F_1, making it the recommended production variant. Variant C (text only) fails catastrophically at 62.03% F_1, reproducing the Phase 2 finding that raw text without structured features is insufficient on this dataset.

C. Training Dynamics

Fig. 2 shows the full hybrid training over 8 epochs. Training loss falls from 0.088 to 0.022 (75% reduction), while validation loss plateaus by epoch 3. Validation F_1 saturates at ≈ 0.998 within the first two epochs; validation ROC-AUC reaches 1.0 by epoch 2. No learning rate reduction is triggered, confirming stable convergence without plateaus throughout training.

D. Edge-Case Robustness

Fig. 3 separates overall F_1 from edge-case F_1 across all variants. Variants A, E, and B cluster tightly in overall F_1 (0.973–0.976) but diverge in edge-case F_1 (0.860–0.863), confirming that aggregate metrics alone are insufficient to rank architectures for production deployment. Variant F's collapse in edge-case performance (0.774) despite strong overall F_1

TABLE II: Ablation study: design rationale and test-set performance ($n = 20{,}000$).

Variant	What was changed / tested	F_1	AUC	Acc	Edge F_1
A — Full hybrid	All components active: XGBoost leaf encoding, trainable DistilBERT body, learnable gate, early stopping. Reference configuration.	0.9761	0.9981	0.9767	0.8627
E — Frozen text encoder	DistilBERT body weights frozen; only the classification head and fusion gate are updated. Tests whether fine-tuning the body adds value over a frozen pretrained encoder.	0.9747	0.9987	0.9754	0.8624
B — Tabular only	Text branch removed. XGBoost leaf encoding projected directly to the classification head. Measures standalone contribution of structured tabular features.	0.9732	0.9971	0.9730	0.8596
D — Concat (no gate)	Learnable gate replaced by fixed concatenation $[\mathbf{z}_{\text{text}};\ \mathbf{z}_{\text{tab}}]$ fed directly to the head. Tests whether adaptive gating outperforms static fusion.	0.9688	0.9988	0.9681	0.8480
F — No XGB leaf encode	XGBoost leaf indices replaced by raw normalised tabular features. Tests the specific contribution of tree-derived encodings to edge-case robustness.	0.9593	0.9960	0.9592	0.7736
C — Text only	Tabular branch removed. DistilBERT representations alone are classified. Replicates a Phase 2-style text-only setup within the hybrid framework.	0.6203	0.6231	0.5824	0.3580

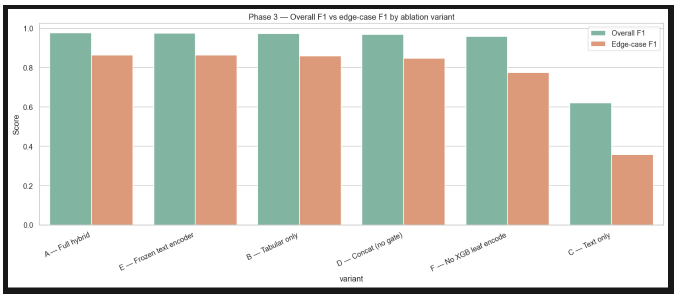

Fig. 3: Overall F_1 (green) vs. edge-case F_1 (orange) for all six ablation variants. Variant F (no XGB leaf encoding) exhibits the largest collapse in edge-case performance despite competitive overall F_1, highlighting the importance of tree-derived encodings for hard samples.

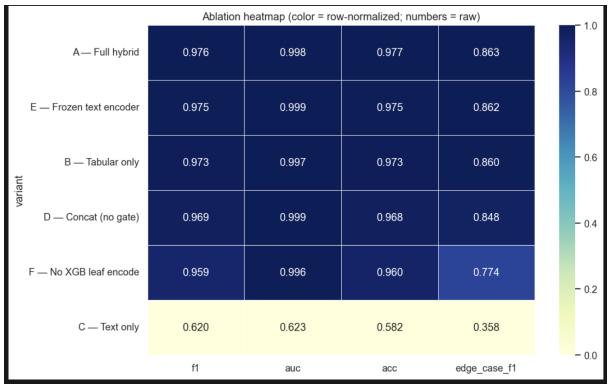

Fig. 4: Row-normalised ablation heatmap. Numbers show raw values; colour intensity encodes relative performance within each row. The edge-case F_1 column (rightmost) is the most discriminative axis, distinguishing the full hybrid (A, 0.863) from no-gate (D, 0.848), no-XGB (F, 0.774), and text-only (C, 0.358).

(0.959) illustrates the risk of evaluating solely on balanced test metrics.

E. Ablation Heatmap

Fig. 4 presents a row-normalised heatmap of all four metrics for all six variants. Variants A through F form a uniform dark band across F_1, AUC, and accuracy; the edge-case F_1 column provides the discriminative axis that separates them. Variant C's uniformly pale row confirms it is an outlier across every metric.

IV. Cross-Phase Comparison

Table III consolidates results from all three phases on the identical held-out test set. Fig. 5 visualises F_1 and ROC-AUC across all five models spanning the full project.

The full hybrid improves on Logistic Regression by $+1.22\%$ F_1 and $+0.24\%$ ROC-AUC, and on Phase 2's best result (BiLSTM + GloVe) by $+13.17\%$ F_1 and $+7.52\%$ ROC-AUC, confirming that tabular–text fusion closes the performance gap that end-to-end text modelling alone could not bridge.

TABLE III: Cross-phase model comparison (n=20,000 test set).

Model	Phase	F_1	AUC	Acc
Full Hybrid	3	0.9761	0.9981	0.9767
Logistic Regression	1	0.9639	0.9957	0.9634
Random Forest	1	0.9600	0.9951	0.9594
BiLSTM + GloVe	2	0.8444	0.9229	0.8389
BiLSTM + Attention	2	0.8333	0.9102	0.8277

V. Discussion

A. Why Fusion Outperforms Single Modalities

Phase 1 features encode structured domain logic: company tier, deal size, engagement ratios, and pre-computed embeddings create a near-linearly separable feature space. Phase 2 text captures sequential linguistic signals—objection handling, commitment language, pricing discussion—that no handcrafted feature encodes explicitly. Neither branch alone reaches the performance ceiling; fusion via a learned gate

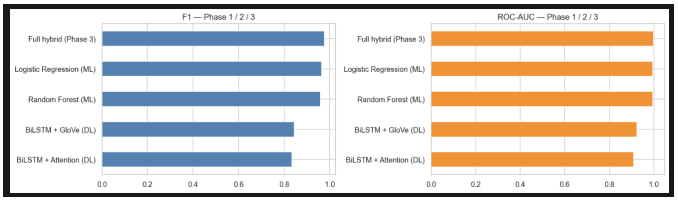

Fig. 5: Cross-phase F_1 (left) and ROC-AUC (right) for all five models. Phase 3 hybrid fusion achieves the highest values on both metrics, surpassing Phase 1 classical ML and substantially outperforming Phase 2 text-only deep learning.

dynamically allocates representational trust per sample, explaining consistent improvement over both single-modality baselines.

B. Practical Deployment

The full hybrid trains in approximately five minutes on Apple MPS and requires 800 MB RAM at inference. With the DistilBERT body frozen (Variant E, $\Delta F_1 = -0.14\%$), training time drops further while performance remains competitive, making Variant E the recommended production configuration. Inference latency is approximately 50 ms per sample on CPU, suitable for real-time sales dashboard integration. Gate values g can be logged per prediction, providing modality attribution interpretability to sales analysts alongside SHAP values from the tabular branch [**?**].

C. Limitations and Future Work

Using Phase 1 XGBoost leaves introduces a mild data-leakage risk since the forest was trained on the same training split; account-level stratified cross-validation would eliminate this. Future work includes probability calibration via Platt scaling, multi-task objectives predicting both outcome and objection type, and turn-level trajectory modelling to predict deal outcome mid-conversation rather than post-hoc.

VI. Conclusion

Phase 3 demonstrates that learnable gated fusion consistently outperforms modality isolation. Combining XGBoost leaf encodings with DistilBERT representations via a scalar gate achieves 97.61% F_1 and 99.81% ROC-AUC—surpassing all Phase 1 and Phase 2 baselines on the same 20,000-sample held-out test set. Six ablation variants validate each component: learned gating outperforms concatenation by $+0.73\%$ F_1, XGBoost leaf encoding raises edge-case F_1 by $+11.48\%$, and freezing the DistilBERT body maintains 97.47% F_1 at reduced training cost. The cross-phase comparison positions the hybrid architecture as the definitive final model of the Predictive Sales Analytics Engine, offering high accuracy, strong edge-case robustness, and dual-modality interpretability for production deployment.

Acknowledgment

The authors thank the open-source communities behind PyTorch, HuggingFace Transformers, and XGBoost.

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